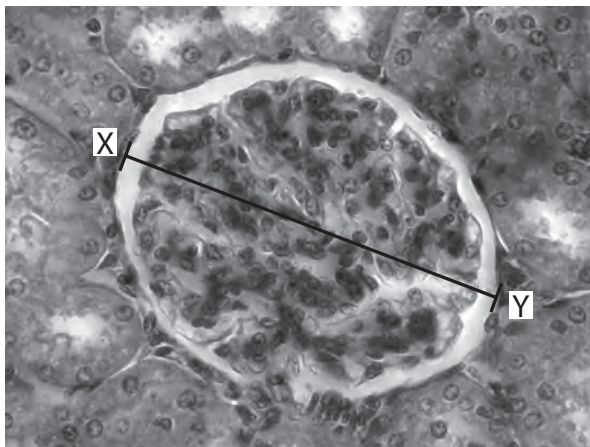


- 6 The kidney has an important role in the regulation of the plasma concentration and blood volume.

The photograph shows a Bowman's capsule (renal capsule).

The line from X to Y shows the diameter of the Bowman's capsule.



(Source: © Science Photo Library/Alamy Stock Photo)

- (a) The diameter from X to Y is 70 nm.
- (i) Calculate the magnification of this photograph.

Give your answer in **standard form**.

(2)

Answer

(ii) Calculate the volume of this Bowman's capsule.

Assume that this Bowman's capsule is spherical.

Use the equation:

$$V = \frac{4}{3}\pi r^3$$

Give your answer to **two** significant figures.

(2)

Answer..... nm³

(b) (i) Which is the specialised receptor that detects changes in the plasma solute concentration?

(1)

- ☐ **A** baroreceptor
- ☐ **B** chemoreceptor
- ☐ **C** osmoreceptor
- ☐ **D** thermoreceptor

(ii) Suggest **one** reason why the plasma solute concentration of the blood needs to be kept within a narrow range.

(1)

(c) Drinking too much water causes the plasma solute concentration to decrease and the blood volume to increase.

Describe how the body responds to this change.

(4)

(d) (i) Which method reabsorbs glucose from the proximal convoluted tubule into the blood?

(1)

- ☐ **A** co-transport with sodium ions
- ☐ **B** endocytosis
- ☐ **C** exocytosis
- ☐ **D** osmosis

- (ii) The Bowman's capsule of a healthy kidney filters up to 7.5 g of glucose per hour.

Calculate the rate of glucose filtration in **milligrams per second**.

(2)

Answer mg s^{-1}

(Total for Question 6 = 13 marks)